NETWORK FORENSICS

Weekly task list

A fictional training network provides flow metadata and DNS events. All addresses are documentation ranges and domains use .example. No live capture or scanning is required.

The investigation question

Which observed flow sequences deserve explanation, and what can metadata not establish?

Your four weekly tasks

Week 1 | Network evidence inventory

Inventory flows.csv and dns.csv, hash both and document collection assumptions. Draw a topology using the two client addresses and named resolver; distinguish source/destination fields from device ownership.

Output: network_map.pdf, field_dictionary.csv and evidence_register.csv

Week 2 | Baseline and outlier description

Group the supplied flows by source, destination and service; calculate event count and bytes for each group. Describe the largest transfer relative to the rest of this small dataset; do not call it exfiltration based solely on size.

Output: flow_summary.csv, dns_mapping.csv and interpretation_notes.pdf

Week 3 | Flow and DNS timeline

Merge DNS and flow events into a UTC timeline with original record IDs. Select one unusual sequence and compare a routine backup explanation with an unverified transfer explanation.

Output: network_timeline.csv, hypothesis_matrix.pdf and evidence_gaps.pdf

Week 4 | Network investigation report

Prepare a 4-6 page report with a topology, aggregate tables, one chart and the selected event sequence. Cite row IDs for every finding and distinguish measured values from assumptions.

Output: network_report.pdf, analysis_tables.csv and reproducibility_notes.pdf

Read pages 2-5 for the full checklists. Submission rules follow on page 6; requirements and installation are on pages 7-8. Worked methods and three reading downloads are on page 9. The complete roadmap is the final page.

WEEK 1 | TASK INSTRUCTIONS

Network evidence inventory

Objective

Create a trustworthy baseline before interpreting any record.

Do this in order

1. Inventory flows.csv and dns.csv, hash both and document collection assumptions.

2. Draw a topology using the two client addresses and named resolver; distinguish source/destination fields from device ownership.

3. Create a field dictionary including timestamp timezone, ports, byte counts and direction. State that packet payloads are not included.

4. Create a case note naming flows.csv and dns.csv. For each, record evidence ID, original filename, byte size, SHA-256 and where the PFSA ZIP was downloaded. Keep the supplied manifest with the ZIP.

5. Define bytes, source address, destination address, destination port and time before using them. A destination port is not proof of the application used.

6. Recheck one result independently: return to the source, repeat the calculation or lookup, and record whether it agrees. Keep a short correction log instead of silently revising previous findings.

Required hand-in

network_map.pdf, field_dictionary.csv and evidence_register.csv

Name the upload PFSA_<student-id>_network_W1.pdf or .zip. A ZIP may contain the report plus CSV, SQL, images or other permitted supporting work. Do not rename a non-PDF file to .pdf. Keep the one portal upload below 10 MB.

Portal description and acceptance check

Write at least 200 genuine words in the description box: what you examined, how you processed it, exact evidence references, results, uncertainty and what you learned. Before submitting, open the uploaded work locally, confirm all filenames resolve, and compare original-file hashes with your intake record.

WEEK 2 | TASK INSTRUCTIONS

Baseline and outlier description

Objective

Turn source records into a documented examination worksheet.

Do this in order

1. Group the supplied flows by source, destination and service; calculate event count and bytes for each group.

2. Describe the largest transfer relative to the rest of this small dataset; do not call it exfiltration based solely on size.

3. Match DNS responses to flows within a documented time window, marking multiple possible matches and missing queries.

4. Keep all original columns from flows.csv and dns.csv. Add new analysis columns rather than replacing source identifiers or timestamps. Preserve the query, formula or filter used.

5. Calculate grouped counts and byte totals on the working copy. Label the sample as a small training dataset, not a normal baseline for a real enterprise.

6. Recheck one result independently: return to the source, repeat the calculation or lookup, and record whether it agrees. Keep a short correction log instead of silently revising previous findings.

Required hand-in

flow_summary.csv, dns_mapping.csv and interpretation_notes.pdf

Name the upload PFSA_<student-id>_network_W2.pdf or .zip. A ZIP may contain the report plus CSV, SQL, images or other permitted supporting work. Do not rename a non-PDF file to .pdf. Keep the one portal upload below 10 MB.

Portal description and acceptance check

Write at least 200 genuine words in the description box: what you examined, how you processed it, exact evidence references, results, uncertainty and what you learned. Before submitting, open the uploaded work locally, confirm all filenames resolve, and compare original-file hashes with your intake record.

WEEK 3 | TASK INSTRUCTIONS

Flow and DNS timeline

Objective

Correlate events without hiding uncertainty or alternative explanations.

Do this in order

1. Merge DNS and flow events into a UTC timeline with original record IDs.

2. Select one unusual sequence and compare a routine backup explanation with an unverified transfer explanation.

3. List the packet, endpoint and application evidence needed to resolve uncertainty; identify what metadata alone cannot reveal.

4. Give every derived timeline event an exact source file and row or event ID. Record raw time, UTC time, conversion rule, evidence meaning and uncertainty in separate fields.

5. Choose a documented time window for DNS matching; retain ambiguous matches, different client addresses and unmatched flows.

6. Recheck one result independently: return to the source, repeat the calculation or lookup, and record whether it agrees. Keep a short correction log instead of silently revising previous findings.

Required hand-in

network_timeline.csv, hypothesis_matrix.pdf and evidence_gaps.pdf

Name the upload PFSA_<student-id>_network_W3.pdf or .zip. A ZIP may contain the report plus CSV, SQL, images or other permitted supporting work. Do not rename a non-PDF file to .pdf. Keep the one portal upload below 10 MB.

Portal description and acceptance check

Write at least 200 genuine words in the description box: what you examined, how you processed it, exact evidence references, results, uncertainty and what you learned. Before submitting, open the uploaded work locally, confirm all filenames resolve, and compare original-file hashes with your intake record.

WEEK 4 | TASK INSTRUCTIONS

Network investigation report

Objective

Deliver a clear, reproducible report that answers the case question.

Do this in order

1. Prepare a 4-6 page report with a topology, aggregate tables, one chart and the selected event sequence.

2. Cite row IDs for every finding and distinguish measured values from assumptions.

3. Provide reproducible grouping/filter criteria and integrity checks so a reviewer can rebuild the results.

4. Write an executive summary first, then scope, evidence register, tool versions, reproducible methods, findings, alternatives, limitations and conclusion. Put detailed tables and queries in appendices.

5. Compare an ordinary backup explanation with an unverified transfer explanation. Large byte counts alone do not establish exfiltration.

6. Recheck one result independently: return to the source, repeat the calculation or lookup, and record whether it agrees. Keep a short correction log instead of silently revising previous findings.

Required hand-in

network_report.pdf, analysis_tables.csv and reproducibility_notes.pdf

Name the upload PFSA_<student-id>_network_W4.pdf or .zip. A ZIP may contain the report plus CSV, SQL, images or other permitted supporting work. Do not rename a non-PDF file to .pdf. Keep the one portal upload below 10 MB.

Portal description and acceptance check

Write at least 200 genuine words in the description box: what you examined, how you processed it, exact evidence references, results, uncertainty and what you learned. Before submitting, open the uploaded work locally, confirm all filenames resolve, and compare original-file hashes with your intake record.

AFTER THE WEEKLY TASK LIST

Submission & assessment

What to submit each week

Submit one main project file and a separate portal narrative of at least 200 words. Use a PDF for a readable report; use a ZIP when the report needs supporting CSV, SQL, JSON, screenshots or source code. The portal supports multiple work formats, but it is not a substitute for organising your evidence.

Item	Required content
Main report	Student ID, domain, week, case question, scope, method, findings and limitations.
Appendices	Source references, derived tables, saved formulas/queries, tool versions and relevant screenshots.
Integrity record	Original file hashes and verification result; explain whether any working file changed.
Portal narrative	At least 200 words written in your own words; a report attachment does not replace this field.

A useful 200-word narrative structure

Use approximately 40-60 words for scope and inputs, 60-90 for your actual method, 60-90 for supported findings, and 40-60 for uncertainty and learning. These are writing prompts, not text to copy. Mention at least two precise record references and one limit of the evidence. Do not repeat a sentence or paste filler just to reach the counter.

Marking rubric | 100 marks per week

Criterion	Marks	What a reviewer looks for
Integrity and scope	20	Correct baseline, preserved originals and a defined question.
Method	25	Reproducible steps, versions, calculations and saved filters.
Analysis	25	Supported reasoning and alternatives, with appropriate uncertainty.
Evidence references	20	Exact file and row/event references for each significant finding.
Communication	10	Clear report, organised package and meaningful portal description.

Calendar deadlines and any resubmission instructions are published in the portal by staff. Four submissions unlock the next document stage; payment confirmation, project assessment and document publication remain staff-controlled. Ask Support about an incorrect upload or locked record.

LAPTOP, SOFTWARE & FILES

Requirements before you begin

Laptop and workspace

Use a laptop or desktop with a supported Windows, macOS or Linux installation, a keyboard and permission to install the required applications. For these small training exports, 4 GB RAM is a practical minimum; 8 GB is recommended. Reserve at least 2 GB free for tools and working files, with additional space for downloaded books. A phone can read PDFs and check portal status, but a laptop is strongly recommended for evidence tables, file packaging and reproducible work.

Required software for this domain

LibreOffice Calc and Writer are required. Wireshark is optional for reading its user guide and future authorised offline-PCAP work; the PFSA pack contains CSV flow/DNS exports, not a packet capture. Python 3 is optional.

Common tools and their purpose

Tool	Use in the assignment
LibreOffice Calc	Import CSVs, retain text identifiers, calculate summaries and save derived tables.
LibreOffice Writer	Write the case report and export it as a readable PDF.
Plain-text editor / VS Code	Inspect JSON, JSONL, SQL and notes without altering formatting in a word processor.
Native SHA-256 utility	Create and compare an integrity baseline for each source file.
Browser and PDF reader	Read the guide, open references and submit work through the student portal.

Training files and folder structure

Download network-training-data.zip from the matching domain card. Its central source files are flows.csv and dns.csv. Save the original ZIP, extract it once, and create four folders: originals, working, analysis and submission. Put the original extracted files in originals, analysis copies in working, derived outputs in analysis, and only final hand-in material in submission.

Before the first task

Record your operating system and the installed application versions. Confirm that the file extensions are visible and that you can create a CSV, open a PDF and make a ZIP. Avoid syncing personal or confidential data into this case folder. The training pack is fictional; no purchase, live investigation account or collection from another person is needed.

PRACTICAL SETUP | WINDOWS, MACOS & LINUX

Installation & first-run checks

1 | Install a document and spreadsheet suite

Open the official LibreOffice download page and choose your operating system and CPU architecture. On Windows, download the stable main installer, run it and follow the standard setup prompts; open both Calc and Writer after installation. On macOS, choose Intel or Apple Silicon as appropriate, open the DMG and move LibreOffice to Applications. On Linux, prefer the distribution's supported software manager. Record the version in your case notes.

LibreOffice download and installation instructions

2 | Set up a plain-text editor

Download VS Code from its official site if you do not already have a suitable editor. On Windows, the user installer normally avoids a machine-wide installation. Open only your working case folder using File > Open Folder. Inspect JSON or SQL as text, keep automatic file formatting off for evidence copies, and do not add extensions that are not needed. A built-in text editor is sufficient when installing VS Code is not possible.

Official VS Code Windows setup

3 | Verify hashes without an extra package

On Windows, open PowerShell in your working case folder and run Get-FileHash -Algorithm SHA256 .\originals\<filename>. On macOS use shasum -a 256 originals/<filename>. On Linux use sha256sum originals/<filename>. Replace the placeholder with one actual filename. Save each digest with its filename, size, date and tool. Repeating the command against the same preserved file should produce the same digest.

4 | Domain-specific installation and first run

Wireshark is optional because the supplied work uses CSVs. If you install it, use the stable build from the official download page and follow the matching operating-system instructions. A packet-capture driver is unnecessary for this CSV assignment. Open the user guide; do not start a live capture. Ask staff for an authorised offline PCAP before attempting packet-level practice.

Wireshark official downloads and user guide

Installation checks: create a test worksheet, export one PDF, read one source file from a working copy and reproduce a SHA-256 digest. Save a short tools.txt with versions and any differences from these instructions. Official setup pages are the source of truth if a vendor changes a screen label.

DOMAIN-SPECIFIC GUIDANCE

Worked method & reading library

First reproducible examination

Import both CSVs with IP addresses and identifiers preserved as text. Create a pivot table grouped by src, dst and dst_port with count(flow_id) and sum(bytes). Save the pivot settings or describe them exactly. For one flow, list every DNS record within your chosen window and explain which client field agrees or conflicts.

A limit you must state

Wireshark does not turn a flow CSV into the original packets. There is no live sniffing, scanning, interception or traffic decryption task in this pack.

Three books & substantial reference downloads

Use the original publisher or author links below. Reference guides and manuals are labelled separately from textbooks. Older editions provide background; they are not a substitute for current tool documentation. Third-party authors retain their rights and do not endorse PFSA.

1 | An Introduction to Computer Networks

Peter L. Dordal | Release 2.0.11, 2023 | Open textbook

Study IP, DNS, transport protocols and the meaning of flow metadata.

Open authorised PDF / author download page

2 | Wireshark User's Guide

Wireshark contributors / Wireshark Foundation | Maintained online edition | Software reference manual

Read interface and offline-analysis chapters; live capture is outside the supplied CSV lab.

Open authorised PDF / author download page

3 | Guide to Computer Security Log Management

Karen Kent and Murugiah Souppaya / NIST | SP 800-92, 2006 | Log management reference guide

Understand log collection, retention, timestamp context and review limitations.

Open authorised PDF / author download page

Download links checked 13 November 2026. Use the source page if a PDF host changes or restricts automated downloads. These reading files are not required attachments to your weekly submission.

FINAL PAGE | PLAN, PRACTISE, VERIFY

Four-week learning roadmap

Your four-stage learning path

Network map -> grouped flow observations -> DNS/flow chronology -> defensible network report

Use this roadmap to plan the sequence, not to calculate calendar deadlines. Staff publish the actual dates in your portal. Early Start changes when access opens; it does not remove any weekly deliverable or assessment requirement.

Stage	Suggested focus	Checkpoint before moving on
Week 1 | Preserve	Days 1-2: scope and inventory. Days 3-4: integrity and working copy. Days 5-7: review and submission.	Every source has an ID, hash and documented limitation.
Week 2 | Examine	Days 8-9: field definitions. Days 10-11: structured checks. Days 12-14: independent recheck and submission.	Derived tables retain original IDs and reproducible calculations.
Week 3 | Correlate	Days 15-16: time conversion. Days 17-18: source correlation. Days 19-21: alternatives and submission.	Timeline retains raw time and explains ambiguity, not just a neat order.
Week 4 | Communicate	Days 22-23: findings. Days 24-25: report and appendices. Days 26-28: independent review and final hand-in.	A reviewer can repeat the method and trace every significant claim.

Daily working habit

Start by reading your last case note and checking the source hash. Work on one defined question, save the method and result, then write a brief end-of-session note: completed work, decisions, outstanding questions and next step. Keep the final report consistent with the latest version of the analysis tables.

Final readiness gate

Confirm four genuine weekly submissions, complete evidence references, original-file integrity checks and an organised final report. Read staff feedback and resolve requested corrections. Keep a personal copy of the submitted package and the portal acknowledgement. Document eligibility and recommendation remain subject to assessment and the academy's published service terms.

What you should be able to demonstrate

Explain the supplied evidence boundary, reproduce one key finding, compare a competing explanation and describe which additional evidence would improve confidence. That explanation is more valuable than a screenshot of a tool running.

Student portal and Support